

Kellogg's® Math In Action Video Series – Cereal Nutrition Video – Teacher Activity Guide

Overview: Students watch a video filmed on location in the Kellogg's® Test Kitchen to learn how to use ratios, proportions, and unit rates to create the nutrition label for Kellogg's® new cereal.



Materials Needed:

- Cereal Nutrition Video Student Guide Copies
- Yummy Oaties Recipe Worksheet Copies
- Rounding Rules to display or hand out per group
- Cereal boxes (maybe samples)

Group students (2-3 students per group)

1. Given the individual ingredient values, calculate out the total macro and micronutrients. Then, with the total numbers, calculate out what those values would be on a 40g serving size (complete this on your Yummy Oaties Recipe Worksheet).

See the Attached Yummy Oaties Recipe Worksheet

2. The FDA utilizes specific rounding rules for numbers with decimals. Using the rounding rules given by your teacher, establish what the rounded nutrients will be. (complete this on your Yummy Oaties Recipe Worksheet).

Make sure you display or hand out the FDA rounding rules for the students to use. See the Attached Yummy Oaties Recipe Worksheet

3. The nutrition facts panel utilizes Daily Values (DV) percentages to help consumers make informed choices. Using the DVs below, calculate the DV percentages.
 - Total Fat 78g
 - Saturated Fat 20g
 - Cholesterol 300mg
 - Sodium 2300mg
 - Total Carbohydrate 275g
 - Dietary Fiber 28g
 - Added Sugars 50g
 - Protein 50g
 - Vitamin D 20mcg
 - Calcium 1300mg
 - Iron 18mg
 - Potassium 4700mg

See the Attached Yummy Oaties Recipe Worksheet

4. Take your calculated values and place them into the blank nutrition facts panel. Note that trans-fat and total sugars will not have a DV % due to no recommended guidance. Protein will also remain blank due to a different way to calculate DV%.



See the Nutrition Facts Panel below #5

5. Find the total calories by utilizing the rounding rules. If total calories <5 , 0 is declared on the label. If 5-50, it is expressed to the nearest 5 increment and if >50 , it is expressed to the nearest 10 increment. Calculate Total Calories using the information below:

- Fat = 9 calories/g
- Carbohydrates = 4 calories/g
- Protein = 4 calories/g

Nutrition Facts		
Serving Size		1 cup (40g/1.4oz)
Amounts Per Serving		
Calories		
	110	
		% Daily Value*
Total Fat	1.5 g	2 %
Saturated Fat	0 g	0 %
Trans Fat	0 g	
Cholesterol	0 mg	0 %
Sodium	0 mg	0 %
Total Carbohydrate	21 g	8 %
Dietary Fiber	4 g	14 %
Total Sugars	11 g	
Includes	8 g Added Sugars	16 %
Protein	4 g	
Vitamin D	0mcg 0%	• Potassium 80mg 0%
Calcium	10mg 0%	• Iron 1.8mg 0%
* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower based on your calorie needs.		

Tolerance Question:

6. A supplier wants to switch out the kind of natural flavor because of a shortage. The new natural flavor has 6g of sugar/100g and 6g added sugar/100g. Given these values, how will the new flavor impact the overall recipe and label? Will the recipe be within 110% of the label?

90% to 110% = 9.9g and 12.1g (Total Sugar) and 7.2g and 8.8g (Added Sugar)

12g (Total Sugar) is in the range, but 10g (Added Sugar) is higher than 8.8g. The new natural flavor could not be used without having to change the label on the box!



Ingredients (per 100g)	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Total Carbs (g)	Dietary Fiber (g)	Total Sugar (g)	Added Sugar (g)	Protein (g)	Vitamin D (mcg)	Calcium (mg)	Iron (mg)	Potassium (mg)
Whole Grain Oat Flour	2	0.5	0	0	1	28	4	0	0	7	0	22	3	185
Corn Flour	2	0	0	0	0	19	2	0	0	3	0	4.5	0.5	16
Dried Strawberries	0	0	0	0	1	4	4	6	0	0	0	6.5	1	0
Sugar	0	0	0	0	0	0	0	18	18	0	0	0	0	0
Natural Flavor	0	0	0	0	2	2	0	3	3	0	0	0	0	90
Salt	0	0	0	0	3	0	0	0	0	0	0	0	0	0
Cinnamon	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total (100g)	4	0.5	0	0	7	53	10	27	21	10	0	33	4.5	291
Per Serving Unrounded (40g)	1.6	0.2	0	0	2.8	21.2	4	10.8	8.4	4	0	13.2	1.8	116.4
Rounded Values (40g)	1.5	0	0	0	0	21	4	11	8	4	0	10	1.8	115
Percent DV	1.923	0	N/A	0	0	7.636	14.286	N/A	16	N/A	0	0.769	10	2.447
Rounded Percent DV	2	0	N/A	0	0	8	14	N/A	16	N/A	0	0	10	0

	Total Fat	Total Carbs	Protein	Total	Total Rounded
Total Calories (unrounded)	14.4	84.8	16	115.2	110

Extension Question

	Total Sugar (g)	Added Sugar (g)
New Natural Flavor	6	6
Total (100g)	30	24
Per Serving Unrounded (40g)	12	9.6
Rounded Values (40g)	12	10
Percent DV	N/A	20
Rounded Percent DV	N/A	20

FDA Nutrition Rounding Rules

- a. Total Fat, Saturated Fat, and Trans-Fat
 - If the value is $< 0.5\text{g}$, 0 is declared on the label
 - If the value is $0.5\text{-}5\text{g}$, use 0.5g increments rounded to the nearest $\frac{1}{2}$ gram
 - If $>5\text{g}$, it is expressed to the nearest 1g increment
- b. Cholesterol
 - If the value is $<2\text{mg}$, 0 is declared on the label
 - If the value is $2\text{-}5\text{mg}$, it is expressed as $<5\text{mg}$
 - If it $>5\text{mg}$, it is expressed to the nearest 5mg increment
- c. Sodium and Potassium
 - If the value is $<5\text{mg}$, it is labeled as 0
 - If the value is $5\text{-}140\text{mg}$, it is expressed to the nearest 5mg increment
 - If it is $>140\text{mg}$, it is expressed to the nearest 10mg increment
- d. Total Carbohydrates, Dietary Fiber, and Sugar
 - If the value is $<0.5\text{g}$, it is labeled as 0
 - If $0.5\text{-}1\text{g}$, it is expressed as $<1\text{g}$
 - If it is $\geq 1\text{g}$, it is expressed to the nearest 1g increment
- e. Protein
 - If $<0.5\text{g}$, it is labeled as 0
 - If $0.5\text{-}1\text{g}$, it is labeled as $<1\text{g}$
 - If $\geq 1\text{g}$, it is expressed to the nearest 1g increment
- f. Vitamin D
 - Round to the nearest 0.1mcg increment
- g. Calcium
 - Round to the nearest 10mg increment
- h. Iron
 - Round to the nearest 0.1mg increment
- i. Vitamins and Minerals expressed as %DV,
 - If $\leq 10\%$ of RDI, express to the nearest 2% DV increment
 - If $>10 - 50\%$ of RDI, express to the nearest 5% DV increment
 - If $>50\%$ of RDI, express to the nearest 10% DV increment

Daily Recommended Values

Based on a 2000 calorie diet

Total Fat 78g
Saturated Fat 20g
Cholesterol 300mg
Sodium 2300mg
Total Carbohydrate 275g
Dietary Fiber 28g
Added Sugars 50g
Protein 50g
Vitamin D 20mcg
Calcium 1300mg
Iron 18mg
Potassium 4700mg